PYTHON PROBLEM-SOLVING ASSIGNMENT

Sample Input • Sample Output • Pseudocode • Flowcharts

10 Programming Problems

Prepared as an academic reference document. All monetary values, rates, slabs and eligibility rules used in the examples are illustrative assumptions so that each problem has a complete worked sample.

Note: Replace the sample policy values with the exact rules supplied by your teacher or assignment sheet if they differ.

1. University Scholarship Allocation

Assumptions / policy used

Illustrative policy: aggregate marks must be at least 80%; every subject must be at least 70; attendance must be at least 75%; annual family income must not exceed ■3,00,000. Rank eligible applicants by aggregate marks (descending), then attendance (descending), then income (ascending).

Sample Input

```
Number of applicants: 3
Applicant A: marks = 88, subject scores = 82, 91, 85; attendance = 90%; income = ■2,50,000
Applicant B: marks = 88, subject scores = 80, 75, 90; attendance = 95%; income = ■2,00,000
Applicant C: marks = 79, subject scores = 85, 82, 80; attendance = 92%; income = ■1,50,000
```

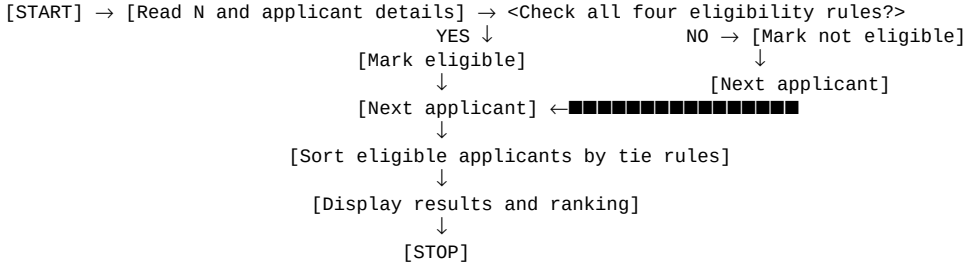
Sample Output

```
A: Eligible
B: Eligible
C: Not eligible (aggregate below 80)
Ranking:
1. Applicant B — 88 marks, 95% attendance, income ■2,00,000
2. Applicant A — 88 marks, 90% attendance, income ■2,50,000
Selected order: B, A
```

Pseudocode

```
START
READ applicant count N
FOR each applicant
    READ name, aggregate, subject scores, attendance, income
    IF aggregate >= 80 AND every subject >= 70
       AND attendance >= 75 AND income <= 300000
        mark applicant ELIGIBLE
    ELSE
        mark applicant NOT ELIGIBLE
    ENDIF
ENDFOR
SORT eligible applicants by:
    aggregate descending, attendance descending,
    income ascending
DISPLAY eligibility and ranked list
STOP
```

Flowchart



2. Progressive Electricity Billing

Assumptions / policy used

Illustrative monthly tariff: first 100 units ■2/unit; next 100 units ■3/unit; units above 200 ■5/unit. Fixed charge ■50/month. Surcharge is 5% of energy charge. Rebate is ■30 when consumption is at most 100 units. An increase is unusual if the current bill is more than 25% above the previous month's bill.

Sample Input

```
Month 1: 80 units
Month 2: 120 units
Month 3: 250 units
```

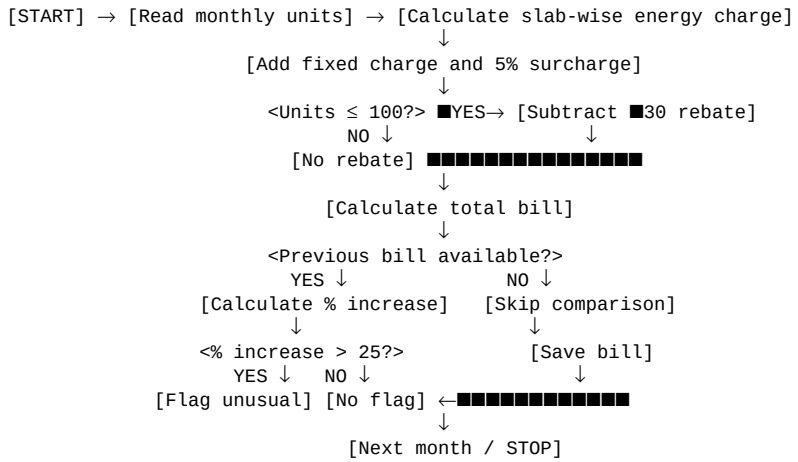
Sample Output

```
Month 1: energy ■160; fixed ■50; surcharge ■8; rebate ■30; total = ■188
Month 2: energy ■260; fixed ■50; surcharge ■13; rebate ■0; total = ■323
Month 3: energy ■650; fixed ■50; surcharge ■32.50; rebate ■0; total = ■732.50
Month 2 increase: 71.8% — unusual
Month 3 increase: 126.9% — unusual
```

Pseudocode

```
START
SET previous_bill = 0
FOR each month
    READ units
    IF units <= 100: energy = units * 2
    ELSE IF units <= 200:
        energy = 100*2 + (units-100)*3
    ELSE:
        energy = 100*2 + 100*3 + (units-200)*5
    fixed = 50
    surcharge = energy * 0.05
    IF units <= 100: rebate = 30 ELSE rebate = 0
    bill = energy + fixed + surcharge - rebate
    DISPLAY bill details
    IF previous_bill > 0:
        increase = (bill-previous_bill)/previous_bill * 100
        IF increase > 25: DISPLAY "Unusual increase"
    previous_bill = bill
ENDFOR
STOP
```

Flowchart



3. Hospital Billing and Patient Classification

Assumptions / policy used

Illustrative policy: add consultation, diagnostic and treatment charges. Senior citizens receive a 10% concession on the total when age is 60 or above. A concession cannot exceed ■2,000. Classify final payment as LOW (below ■5,000), MEDIUM (■5,000–■15,000), or HIGH (above ■15,000).

Sample Input

```
Patient: Ravi
Age: 65
Consultation: ■800
Diagnostics: ■2,200
Treatment: ■9,000
```

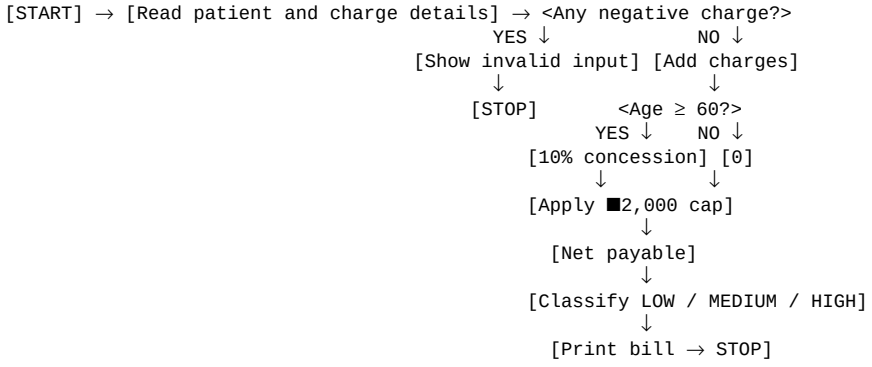
Sample Output

```
Gross bill = 800 + 2,200 + 9,000 = ■12,000
Senior concession = 10% of ■12,000 = ■1,200
Allowed concession = ■1,200 (below ■2,000 cap)
Net payable = ■10,800
Classification = MEDIUM
```

Pseudocode

```
START
READ patient name, age, consultation, diagnostics, treatment
IF any charge < 0: DISPLAY invalid bill; STOP
 gross = consultation + diagnostics + treatment
IF age >= 60: concession = gross * 0.10
ELSE: concession = 0
IF concession > 2000: concession = 2000
net = gross - concession
IF net < 5000: class = LOW
ELSE IF net <= 15000: class = MEDIUM
ELSE: class = HIGH
DISPLAY itemized bill, concession, net, class
STOP
```

Flowchart



4. Employee Payroll and Deduction Analysis

Assumptions / policy used

Illustrative monthly payroll: gross = basic + allowances + overtime + incentives. Employee contribution is 5% of basic pay. Income tax on gross: 0% up to ₦20,000; 5% on the portion from ₦20,001 to ₦40,000; 10% on the portion above ₦40,000. Net salary = gross − all deductions.

Sample Input

```
Basic pay: ₦30,000
Allowances: ₦5,000
Overtime: 10 hours at ₦200/hour
Incentive: ₦2,000
```

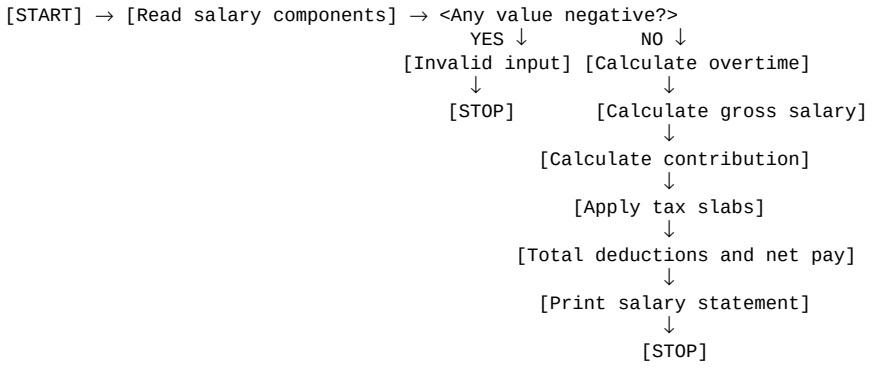
Sample Output

```
Overtime pay = 10 × 200 = ₦2,000
Gross salary = 30,000 + 5,000 + 2,000 + 2,000 = ₦39,000
Employee contribution = 5% of 30,000 = ₦1,500
Income tax = 5% of (39,000 − 20,000) = ₦950
Total deductions = ₦2,450
Net salary = ₦36,550
```

Pseudocode

```
START
READ basic, allowances, overtime_hours, overtime_rate, incentive
IF any input < 0: DISPLAY invalid input; STOP
overtime_pay = overtime_hours * overtime_rate
gross = basic + allowances + overtime_pay + incentive
contribution = basic * 0.05
IF gross <= 20000: tax = 0
ELSE IF gross <= 40000: tax = (gross-20000) * 0.05
ELSE: tax = 20000*0.05 + (gross-40000)*0.10
deductions = contribution + tax
net = gross - deductions
DISPLAY formatted salary statement
STOP
```

Flowchart



5. Multi-Plan Telecommunications Cost Optimization

Assumptions / policy used

Compare plans using monthly usage. A plan is eligible only if its validity is at least the required days and its base price is within the budget. Excess data is charged at the plan's stated rate. Choose the eligible plan with the lowest total cost. Voice/SMS are assumed included for this example.

Sample Input

```
Required validity: 28 days; budget: ■350; data needed: 2.5 GB
Plan A: price ■199, validity 28 days, data 2 GB, excess ■20/GB
Plan B: price ■299, validity 30 days, data 3 GB, excess ■20/GB
Plan C: price ■249, validity 21 days, data 3 GB, excess ■10/GB
```

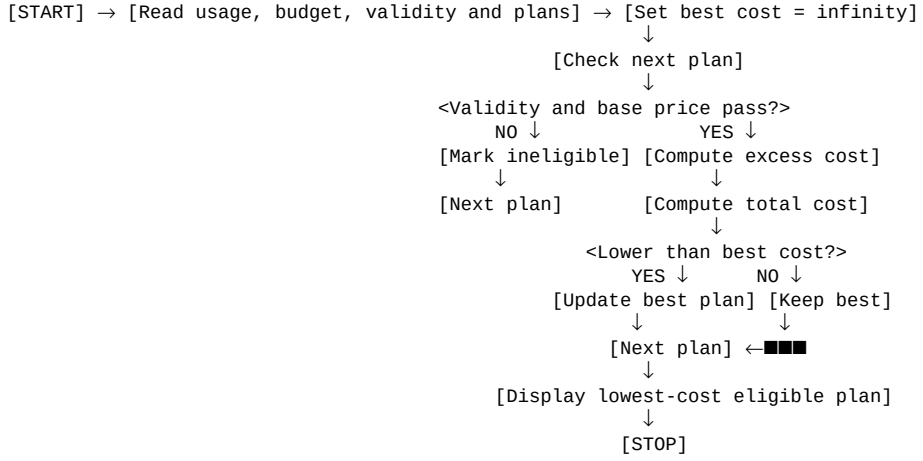
Sample Output

```
Plan A: eligible; excess = 0.5 GB; excess charge = ■10; total = ■209
Plan B: eligible; excess = 0 GB; excess charge = ■0; total = ■299
Plan C: not eligible (validity below 28 days)
Most economical eligible plan: Plan A — ■209
```

Pseudocode

```
START
READ required_days, budget, required_data
READ plan count N
best_plan = NONE; best_cost = infinity
FOR each plan
    READ name, price, validity, included_data, excess_rate
    IF validity >= required_days AND price <= budget
        excess = MAX(0, required_data - included_data)
        total = price + excess * excess_rate
        DISPLAY plan and total
        IF total < best_cost:
            best_cost = total; best_plan = name
    ELSE
        DISPLAY plan not eligible
    ENDIF
ENDFOR
IF best_plan is NONE: DISPLAY no eligible plan
ELSE: DISPLAY best_plan and best_cost
STOP
```

Flowchart



6. Railway Fare and Reservation Calculator

Assumptions / policy used

Illustrative one-way fare: distance up to 100 km costs ■1.50/km; distance above 100 km costs ■1.20/km for the portion above 100. AC class multiplier = 2.0; Sleeper multiplier = 1.0. Passengers aged 60+ receive 40% concession. Reservation fee = ■30 per passenger. Cancellation penalty = 20% of fare before reservation fee. The sample calculates a booking, not a cancellation.

Sample Input

```
Distance: 150 km; class: Sleeper
Passenger 1: age 65
Passenger 2: age 30
Passengers: 2
```

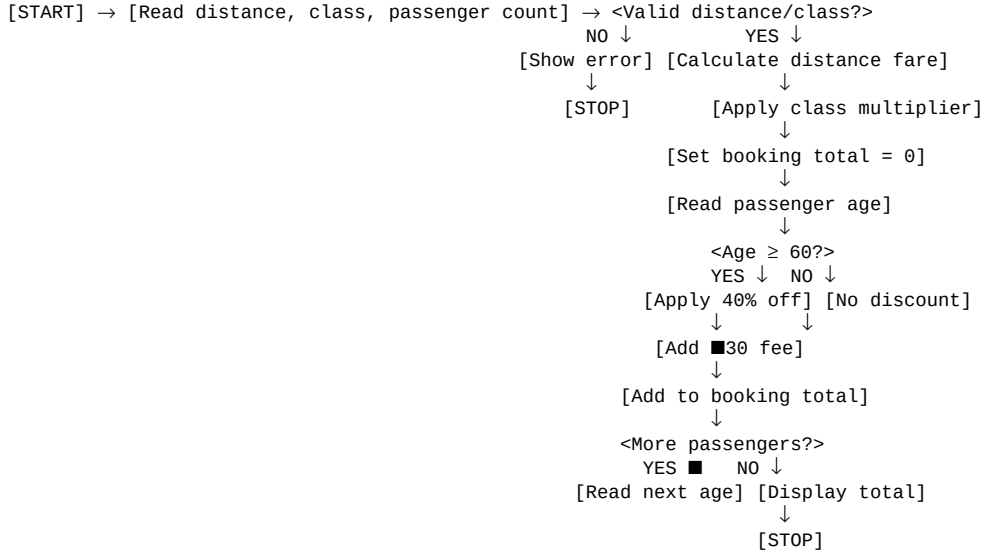
Sample Output

```
Base fare per passenger = 100×1.50 + 50×1.20 = ■210
Passenger 1: concession 40% = ■84; fare after concession ■126; reservation ■30; total ■156
Passenger 2: no concession; fare ■210; reservation ■30; total ■240
Total booking amount = ■396
```

Pseudocode

```
START
READ distance, travel_class, passenger count N
IF distance <= 0: DISPLAY invalid distance; STOP
base = MIN(distance,100)*1.50 + MAX(0,distance-100)*1.20
IF class = AC: base = base * 2.0
ELSE IF class = Sleeper: keep base
ELSE: DISPLAY invalid class; STOP
total = 0
FOR each passenger
    READ age
    fare = base
    IF age >= 60: fare = fare * 0.60
    reservation = 30
    passenger_total = fare + reservation
    DISPLAY passenger_total
    total = total + passenger_total
ENDFOR
DISPLAY total booking amount
STOP
```

Flowchart



7. Retail Billing with Multiple Discount Rules

Assumptions / policy used

Illustrative rules: electronics tax 18%; groceries tax 5%; clothing tax 12%. If pre-tax subtotal is at least ■5,000, apply a 10% discount; otherwise no threshold discount. Maximum total discount is ■1,000. Apply discount before tax. Each line amount = quantity × unit price.

Sample Input

```
Item 1: Headphones, electronics, quantity 1, unit price ■3,000
Item 2: Rice, groceries, quantity 2, unit price ■1,200
Item 3: Shirt, clothing, quantity 1, unit price ■1,000
```

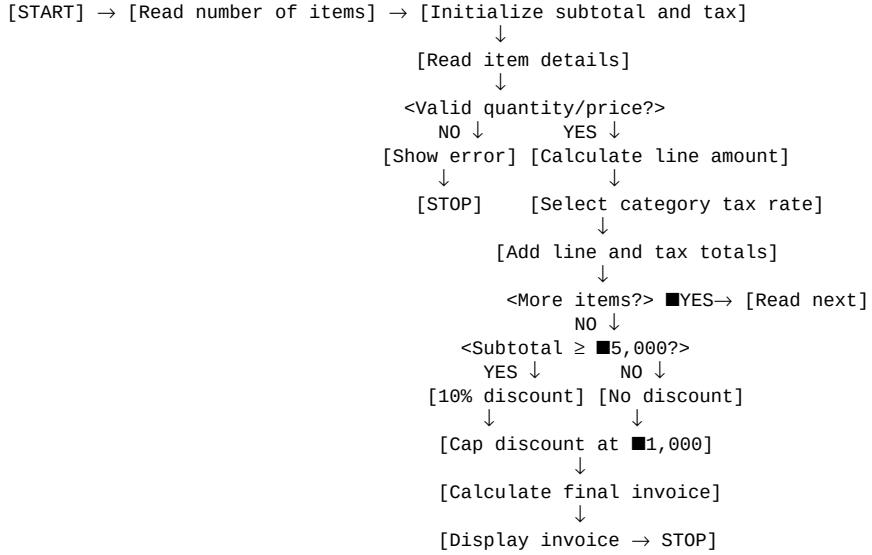
Sample Output

```
Headphones: ■3,000; tax 18% = ■540
Rice: ■2,400; tax 5% = ■120
Shirt: ■1,000; tax 12% = ■120
Subtotal = ■6,400; threshold discount = 10% = ■640 (within ■1,000 cap)
Tax total = ■780; final invoice = 6,400 − 640 + 780 = ■6,540
```

Pseudocode

```
START
READ item count N; subtotal = 0; tax_total = 0
FOR each item
    READ name, category, quantity, unit_price
    IF quantity < 0 OR unit_price < 0: DISPLAY invalid; STOP
    line = quantity * unit_price
    SELECT tax_rate by category:
        electronics 0.18; groceries 0.05; clothing 0.12
    tax = line * tax_rate
    subtotal = subtotal + line
    tax_total = tax_total + tax
    STORE item details
ENDFOR
IF subtotal >= 5000: discount = subtotal * 0.10
ELSE: discount = 0
 discount = MIN(discount, 1000)
final = subtotal - discount + tax_total
DISPLAY itemized invoice and final
STOP
```

Flowchart



8. Manufacturing Cost and Profitability Analysis

Assumptions / policy used

Total material cost = quantity × material cost per unit. Wastage cost is calculated as a percentage of material cost. Total production cost = material + wastage + labour + machine cost. Profit = total sales – total production cost. Target profit margin is 20% of production cost.

Sample Input

```
Units produced: 100
Material cost per unit: ■50
Wastage rate: 5%
Labour cost: ■2,000
Machine cost: ■1,000
Selling price per unit: ■90
```

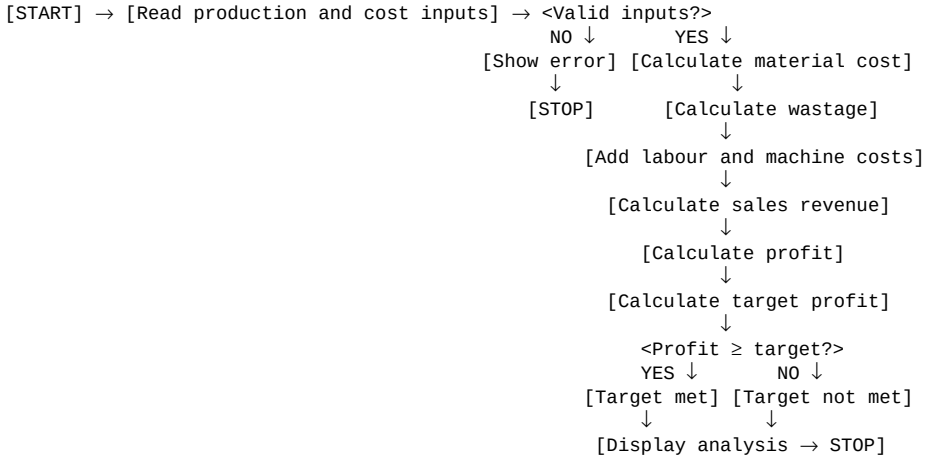
Sample Output

```
Material cost = 100 × ■50 = ■5,000
Wastage cost = 5% × ■5,000 = ■250
Total production cost = 5,000 + 250 + 2,000 + 1,000 = ■8,250
Sales revenue = 100 × ■90 = ■9,000
Profit = ■750; profit margin on cost = 9.09%
Target profit = 20% × ■8,250 = ■1,650
Result: target not met
```

Pseudocode

```
START
READ units, material_rate, wastage_percent, labour, machine, selling_price
IF units <= 0 OR any cost < 0: DISPLAY invalid; STOP
material = units * material_rate
wastage = material * wastage_percent / 100
production_cost = material + wastage + labour + machine
revenue = units * selling_price
profit = revenue - production_cost
target_profit = production_cost * 0.20
IF profit >= target_profit: DISPLAY target met
ELSE: DISPLAY target not met
DISPLAY all cost and profit values
STOP
```

Flowchart



9. Multi-Currency Travel Budget Analysis

Assumptions / policy used

All exchange rates are supplied as local-currency units per 1 unit of foreign currency. Conversion fee is 2% of converted amount. Budget is in Indian rupees. The example uses USD, EUR and GBP expenses; rates are illustrative and are not live exchange rates.

Sample Input

```
Budget: ■1,00,000; conversion fee: 2%
Hotel: USD 300; rate ■83/USD
Food: EUR 200; rate ■90/EUR
Transport: GBP 100; rate ■105/GBP
```

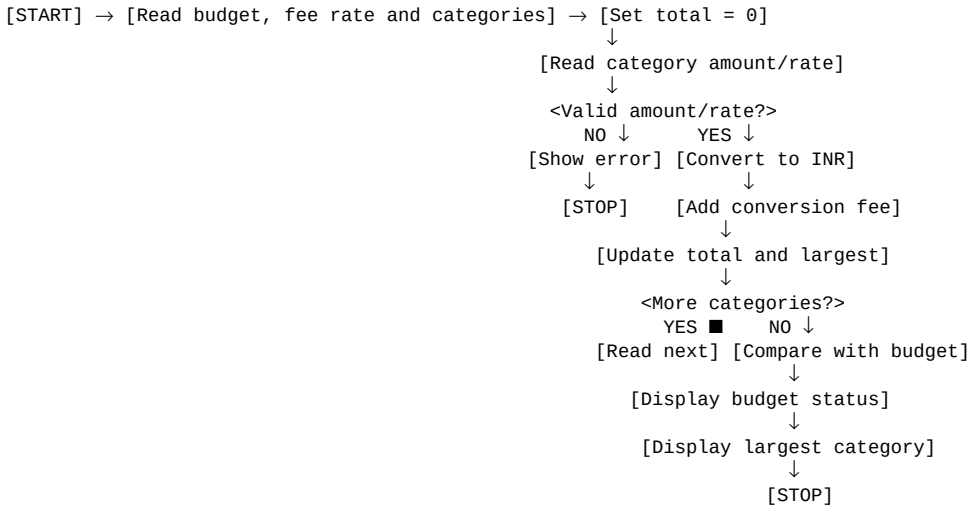
Sample Output

```
Hotel: 300 × 83 = ■24,900; fee ■498; total ■25,398
Food: 200 × 90 = ■18,000; fee ■360; total ■18,360
Transport: 100 × 105 = ■10,500; fee ■210; total ■10,710
Total planned spend = ■54,468
Budget remaining = ■45,532 — within budget
Largest expense category = Hotel (■25,398 including fee)
```

Pseudocode

```
START
READ budget, fee_percent, category count N
spend_total = 0; largest_amount = -1
FOR each category
    READ category, foreign_amount, exchange_rate
    IF amount < 0 OR rate <= 0: DISPLAY invalid; STOP
    converted = foreign_amount * exchange_rate
    fee = converted * fee_percent / 100
    local_total = converted + fee
    spend_total = spend_total + local_total
    IF local_total > largest_amount:
        largest_amount = local_total
        largest_category = category
    DISPLAY category cost
ENDFOR
IF spend_total <= budget: DISPLAY within budget
ELSE: DISPLAY over budget and shortfall
DISPLAY total spend, remaining/shortfall, largest category
STOP
```

Flowchart



10. Campus Event Financial Planning

Assumptions / policy used

Fixed costs are venue, equipment and transport. Variable costs are catering and certificates per participant. Sponsorship is deducted from total event cost. Required registration fee is the remaining cost divided by registered participants. Break-even registration count is the smallest whole number n satisfying n × fee + sponsorship ≥ fixed costs + n × variable cost per participant. Example assumes fee ■500.

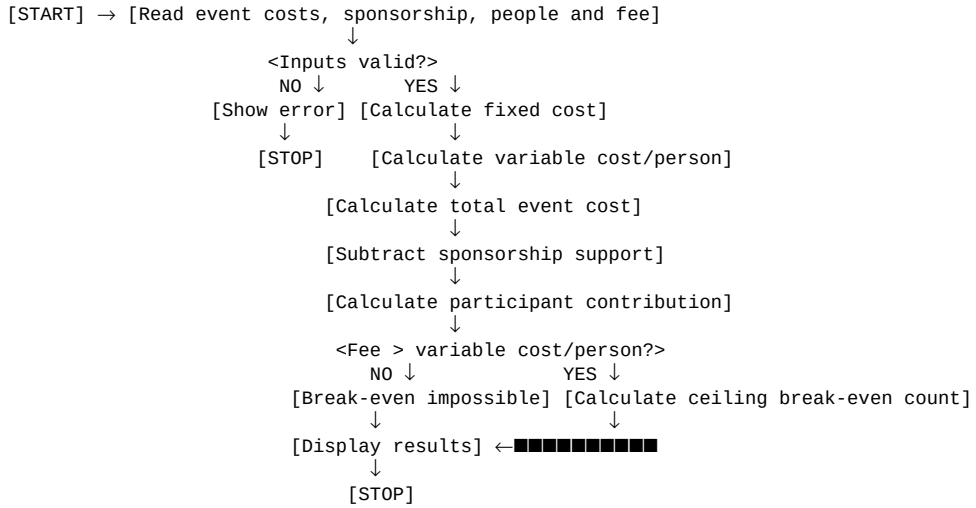
Sample Input

```
Venue: ■10,000; equipment: ■5,000; transport: ■3,000
Catering per participant: ■150; certificate per participant: ■20
Sponsorship: ■8,000; planned participants: 100
Registration fee: ■500
```

Sample Output

```
Fixed cost = 10,000 + 5,000 + 3,000 = ■18,000
Variable cost per participant = 150 + 20 = ■170
Total cost for 100 = 18,000 + (100 × 170) = ■35,000
Amount after sponsorship = 35,000 − 8,000 = ■27,000
Required contribution per participant = ■270
At ■500 fee, contribution after variable cost = ■330/person
Break-even registrations = ceiling((18,000 − 8,000) / (500 − 170)) = 31 participants
```

Pseudocode

```
START
READ venue, equipment, transport, catering_per_person,
     certificate_per_person, sponsorship, planned_people, fee
IF any cost < 0 OR planned_people < 0: DISPLAY invalid; STOP
fixed = venue + equipment + transport
variable = catering_per_person + certificate_per_person
total_cost = fixed + planned_people * variable
net_funding_need = MAX(0, total_cost - sponsorship)
IF planned_people > 0: contribution = net_funding_need / planned_people
ELSE: contribution = 0
margin_per_person = fee - variable
IF margin_per_person <= 0:
    DISPLAY break-even not possible at this fee
ELSE:
    break_even = CEILING(MAX(0, fixed-sponsorship) / margin_per_person)
DISPLAY cost, contribution, sponsorship gap, break_even
STOP
```

Flowchart